Pulse-Waveform Design and Phase-Engineered Follow-Up for Selective Qubit Rotation in Dispersive cQED

March 19, 2026

1 Objective and Scope

This document is a unified report for the `sqr_pulse_waveform_design` study. It has two parts:

1. a reproduced summary of the original study, reconstructed from the saved data, figures, and validation artifacts after the prior manuscript source was overwritten, and

2. a new follow-up focused on the question the original study left open: whether the remaining strict-logical error is mostly a correctable cavity-only Fock-dependent phase profile or a deeper waveform/control limitation.

The baseline study established the envelope hierarchy, the practical operating window, the role of GRAPE as an upper bound, and the qualitative importance of phase structure. The new follow-up turns that diagnosis into a direct compiled control test.

2 Part I: Reproduced Original Study

2.1 System, targets, and waveform families

The original study used a dispersive transmon–cavity model with $\omega_q/2\pi = 6.150\,\mathrm{GHz}$, $\omega_c/2\pi = 5.241\,\mathrm{GHz}$, $\alpha/2\pi = -255\,\mathrm{MHz}$, $\chi/2\pi = -2.84\,\mathrm{MHz}$, a qutrit transmon ($|g\rangle, |e\rangle, |f\rangle$), and a truncated cavity logical window $n = 0, \ldots, 3$. The main scan variable was the dimensionless selectivity parameter $\chi T/2\pi$.

Four baseline families were compared in the closed system:

- single-tone Gaussian,
- square,
- cosine-squared (Hann),
- one-segment multitone with a Gaussian envelope.

The original questions were whether true SQR or conditional-phase SQR is the right primitive, which smooth envelope performs best, where the practical selectivity–decoherence optimum lies, and how far structured parametric pulses sit below the GRAPE upper bound.

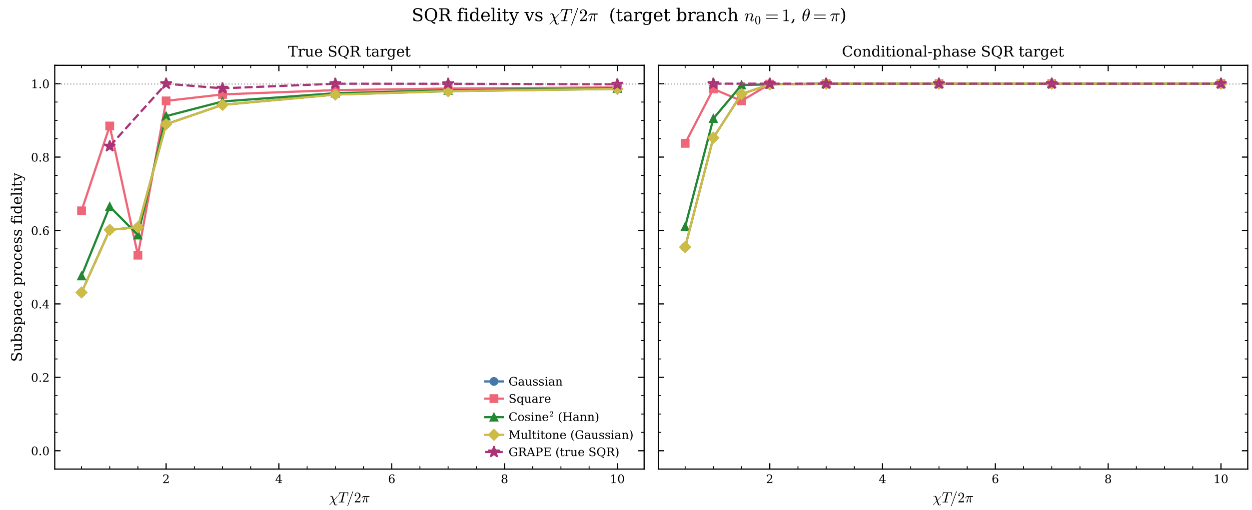

Figure 1: Reproduced baseline closed-system result from the original study. Smooth single-tone envelopes rapidly approach unit conditional-phase SQR fidelity, while true-SQR fidelity remains limited by spectator phases.

2.2 Closed-system baseline conclusions

Figure 1 reproduces the central original result. The conditional-phase target is dramatically easier than true SQR, and smooth single-tone envelopes already reach essentially unit conditional-phase fidelity in the practical regime.

The saved baseline data give the following concrete numbers:

- Gaussian conditional-phase fidelity rises from 0.555 at $\chi T/2\pi = 0.5$ to 0.998 at 2.0 and 0.99999 at 3.0.
- Cosine-squared performs best at intermediate duration, reaching 0.9963 already at $\chi T/2\pi = 1.5$, 0.9981 at 2.0, and 0.999997 at 3.0.
- Leakage across the full baseline scan stays below 7.4×10^{-5}, validating the effective two-level picture for the qubit manifold.

The reproduced message is the same as in the original report: conditional-phase SQR, not strict true SQR, is the natural baseline primitive.

2.3 GRAPE, decoherence, and practical operating point

The original study also established three system-level facts that remain intact:

1. **GRAPE is the proper upper bound.** Saved GRAPE results give conditional-phase fidelity $F_{\text{GRAPE}} \approx 1$ for all scanned $\chi T/2\pi \in \{1, 2, 3, 5, 7, 10\}$, while true-SQR GRAPE is already 0.8295 at $\chi T/2\pi = 1$ and exceeds 0.987 by $\chi T/2\pi = 3$.
2. **The practical open-system optimum is short and smooth.** In the original Phase-5 data, the cosine-squared pulse achieved the best net conditional-phase fidelity, peaking at 0.9845 around $\chi T/2\pi = 1.5$. Gaussian and multitone one-segment baselines peaked near $\chi T/2\pi = 2.0$ at about 0.9814.

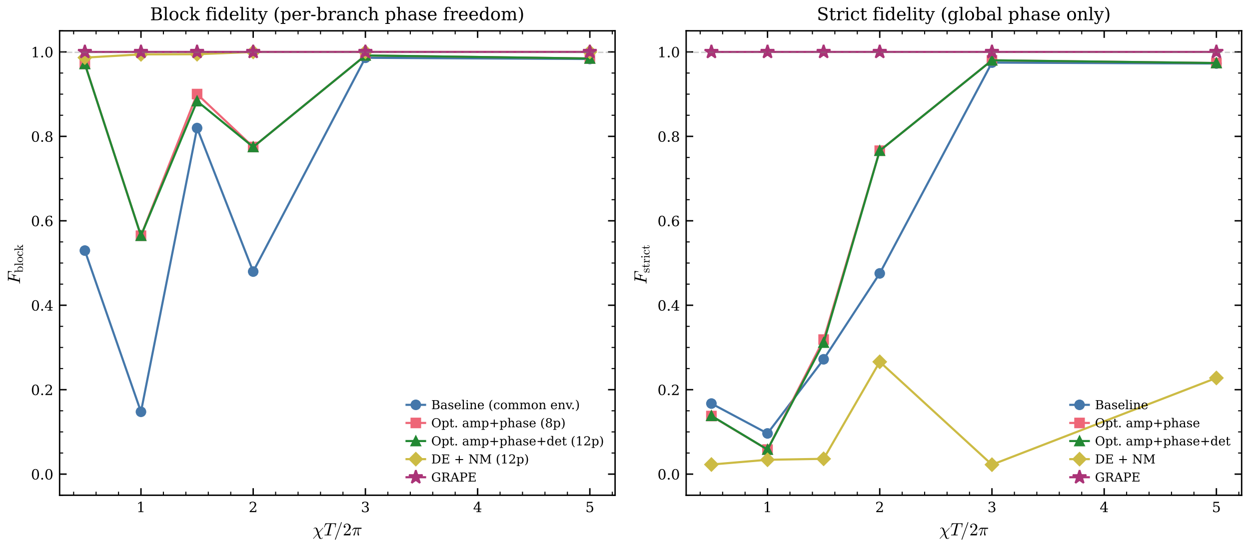

Figure 2: Reproduced original all-branch result. Parametric multitone pulses can be very good under a block-phase-relaxed metric while remaining very far from the strict logical unitary.

3. **The original hardware-constrained GRAPE study did not see a DAC or amplitude-ceiling bottleneck.** The saved hardware-constrained and unconstrained GRAPE conditional-phase fidelities are both numerically 1.0 across the scan.

These three points matter because they show the system is controllable enough for high-fidelity logical action; the main remaining problem is identifying the right *structured* primitive below the GRAPE ceiling.

2.4 Generalized targets, optimized multitone, and all-branch behavior

The original extension phases refined the interpretation of that gap:

- The generalized-target study showed that echoed sequences can improve a block-phase-relaxed metric while hurting strict logical performance via same-block loss and leakage.
- The optimized single-branch multitone study improved the block-relaxed fidelity substantially only at very short gate times; for $\chi T/2\pi \geq 3$ it provided at most a marginal gain over the smooth single-tone baseline.
- The all-branch study produced the sharpest warning sign: DE+NM multitone pulses reached $F_{\text{block}} \geq 0.987$ but only $F_{\text{strict}} \leq 0.27$, while GRAPE stayed at $F \geq 0.9999$.

Figure 2 reproduces that last point.

This all-branch gap motivated the follow-up: was the missing error mostly a cavity-only diagonal phase structure, or something deeper?

3 Part II: Phase-Compilation Follow-Up

3.1 Reframed control objective

The new follow-up was organized around the compiled primitive

$$U_{\text{compiled}} \approx U_{\text{diag}} U_{\text{SQR}}, \qquad U_{\text{diag}} = \sum_n e^{i\phi_n} |n\rangle\langle n|.$$

The key question was whether a cavity-only phase layer, combined with one global qubit virtual-Z, can recover most of the strict-logical gap left by the baseline waveforms.

This defines three nested gauges:

(a) raw strict logical fidelity,

(b) strict fidelity after one global qubit virtual-Z and cavity-only phase compilation,

(c) branch-local-Z-relaxed fidelity, which serves as the upper bound if the remaining defect is mostly per-branch qubit phase rather than bosonic phase.

3.2 Follow-up cases

Two complementary follow-up cases were run.

Single-target enlarged-window case. The standard single-target $R_X(\pi)$ SQR on branch $n_0 = 1$ was re-evaluated on an enlarged logical window $n = 0, \ldots, 7$ ($N = 8$, $n_{\text{cav}} = 10$) for the Gaussian and cosine-squared families over $\chi T/2\pi \in \{1, 2, 3, 5\}$.

Representative short-gate structured all-branch case. The previously defined Phase-B independent-tone all-branch multitone optimizer was re-run at $\chi T/2\pi = 0.5$ and 1.0 to test whether cavity-only compilation can rescue a structured multitone family precisely where the original report suggested phase engineering might matter most.

3.3 Single-target SQR does induce a cavity-only phase profile, and it is almost perfectly linear

Figure 3 shows the extracted cavity-only phase profile for the enlarged-window single-target problem. The phase is not only present; it is extremely smooth. Across both families and all scanned durations, the worst linear-fit RMS phase error is only 4.59×10^{-5} rad.

At $\chi T/2\pi = 3$, both baselines are described to numerical precision by

$$\phi(n) \approx -0.0684\, n \quad \text{rad},$$

after gauge-fixing $\phi(0) = 0$.

This is the strongest positive answer produced by the follow-up: in the single-target setting, the cavity-only phase profile is real, low-complexity, and naturally compilable.

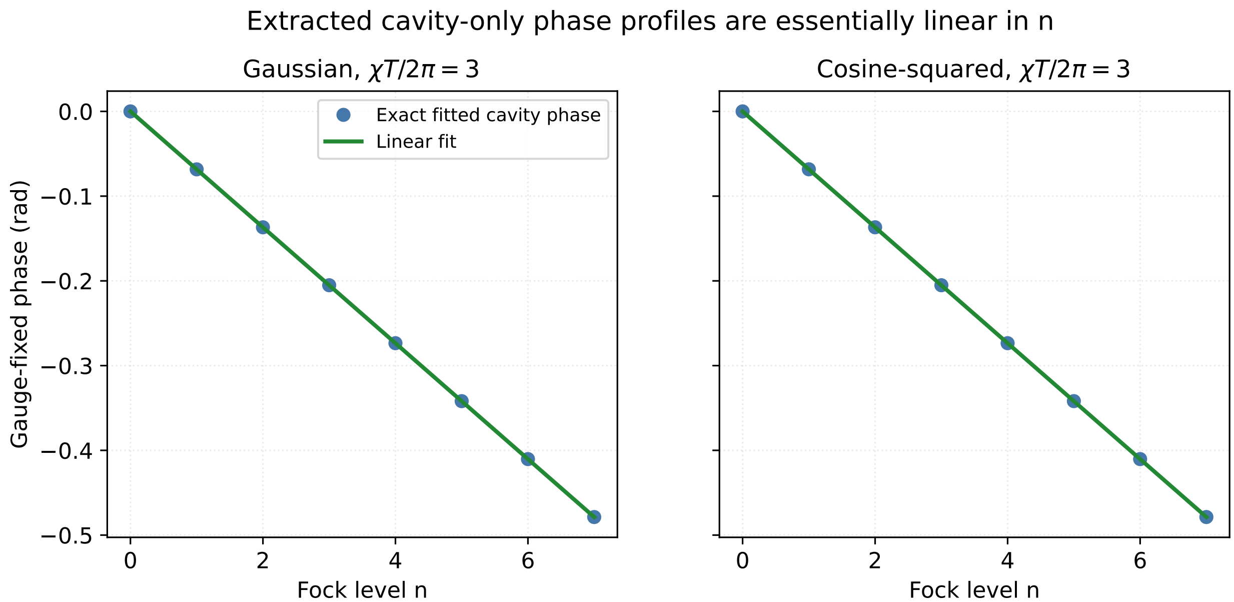

Figure 3: Follow-up phase extraction on the enlarged $N = 8$ logical window. The fitted cavity-only phase profile is essentially linear in photon number for both Gaussian and cosine-squared single-target SQR.

3.4 Cavity-only compilation helps on larger logical windows, but it does not close the full gap

Figure 4 summarizes the single-target compiled-control comparison.

The concrete improvements are:

- Gaussian: $0.738 \to 0.755$ at $\chi T/2\pi = 1$ and $0.927 \to 0.949$ at $\chi T/2\pi = 3$.
- Cosine-squared: $0.777 \to 0.795$ at $\chi T/2\pi = 1$ and $0.931 \to 0.954$ at $\chi T/2\pi = 3$.

The gain is small but systematic. The minimum improvement over the scan is 0.0167, and for the practical regime $\chi T/2\pi \geq 2$ it remains above 0.0217 for both families.

However, cavity-only compilation does *not* recover the full gap to the branch-local-Z-relaxed fidelity. At $\chi T/2\pi = 3$, the cosine-squared pulse reaches $F_{\mathrm{diag}} = 0.954$ after cavity-only compilation, while the branch-local-Z-relaxed upper bound is already 0.999998.

So the single-target follow-up gives a nuanced answer:

> The strict-logical deficit is partly a cavity-phase problem, but not purely a cavity-phase problem.

3.5 Coherent superposition benchmarks improve after cavity-phase compilation

The enlarged-window coherence benchmark confirms that the compiled phase layer is not just improving an operator-overlap metric. At $\chi T/2\pi = 3$:

- Gaussian pair-superposition fidelity improves from mean/min 0.959/0.671 to 0.974/0.868.
- Cosine-squared improves from mean/min 0.961/0.686 to 0.976/0.879.

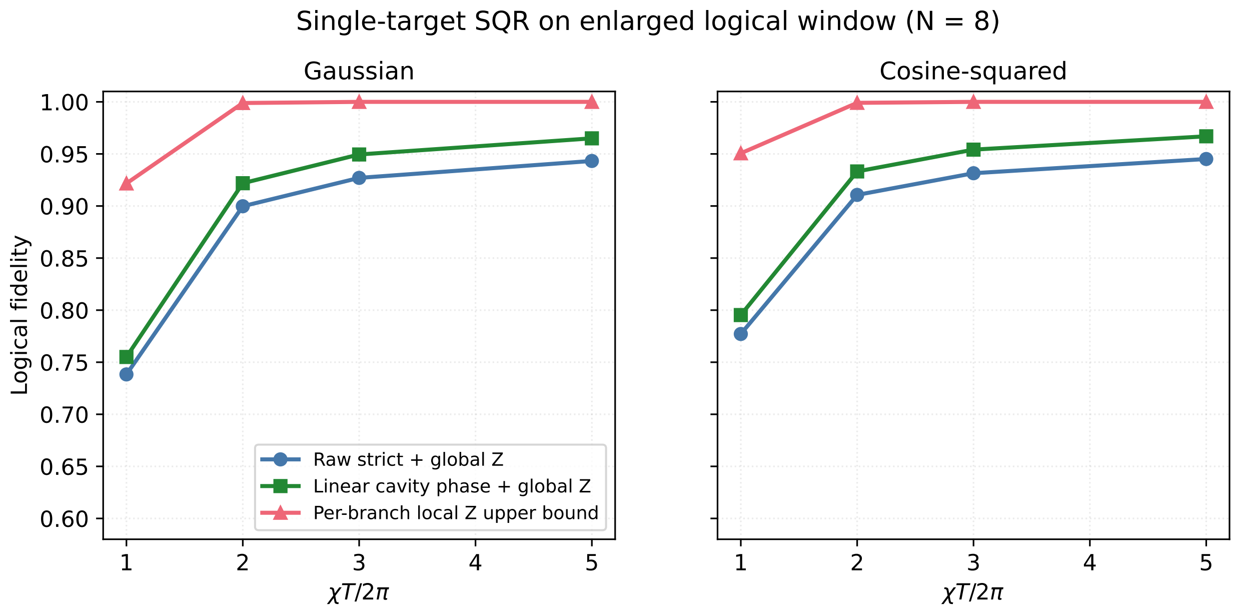

Figure 4: Follow-up compiled comparison on the enlarged $N = 8$ logical window. A linear cavity-only phase layer improves the strict logical action, but does not reach the branch-local-Z-relaxed upper bound.

3.6 The representative all-branch short-gate problem is not a cavity-only phase-compilation problem

The same conclusion does not hold for the structured all-branch short-gate case. The representative Phase-B independent-tone optimizer gives:

$\chi T/2\pi$	F_{raw}	F_{diag}	F_{branchZ}	F_{GRAPE}
0.5	0.138	0.028	0.972	0.9999
1.0	0.058	0.059	0.565	1.0000

At $\chi T/2\pi = 0.5$, the fitted cavity-only phase profile is irregular,

$$\phi \approx (0,\ 1.50,\ -0.14,\ 1.36)\ \mathrm{rad},$$

and applying it actually *worsens* the strict logical fidelity. This means the structured all-branch short-gate failure mode is not a missing bosonic SNAP-like layer; it is a deeper branch-local qubit-phase and/or non-diagonal control problem.

4 Unified Interpretation

The combined baseline-plus-follow-up picture is now much clearer.

What the original study already proved. Smooth single-tone selective pulses, especially cosine-squared, are already excellent conditional-phase SQR primitives in the practical operating regime. GRAPE demonstrates that the system itself is not the bottleneck. The all-branch extension showed that phase structure matters and that block-relaxed success can severely overestimate strict logical quality.

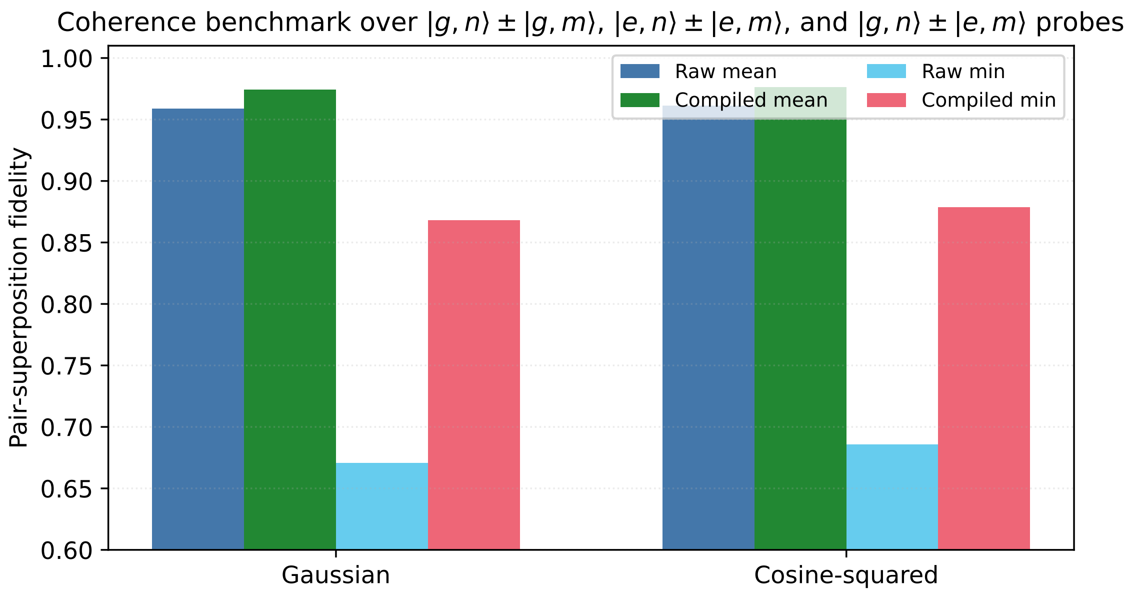

Figure 5: Follow-up coherent-superposition benchmark at $\chi T/2\pi = 3$ on the enlarged $N = 8$ logical window. A cavity-only compiled phase layer improves both the mean and worst-case probe fidelity.

What the follow-up adds. The missing strict-logical error is not of a single type.

- For single-target SQR on larger logical windows, there is a real and almost perfectly linear cavity-only phase profile. Compiling it helps.
- But even there, the residual gap to the branch-local-Z limit is substantial, so cavity-only compilation is not the whole story.
- For structured all-branch short-gate multitone control, cavity-only compilation is the wrong correction class entirely.

So the strongest version of the hypothesis "the remaining strict-logical error is mostly a bosonic diagonal phase" is not supported uniformly across the study space. The correct statement is narrower and more useful:

> Low-order cavity-phase compilation is a real, calibratable improvement for single-target SQR on larger logical windows, but the hardest structured multitone failures are not bosonic-phase problems.

5 Practical Recommendation

The unified experiment-facing recommendation is summarized in Table 1.

The practical decision is therefore:

1. Use cosine-squared single-target SQR as the default experiment-facing primitive in the practical $\chi T/2\pi \sim 2$–3 regime.
2. When logical windows are enlarged and inter-Fock coherence matters, add a low-order cavity-only phase compilation layer; it is simple, low-complexity, and demonstrably helpful.

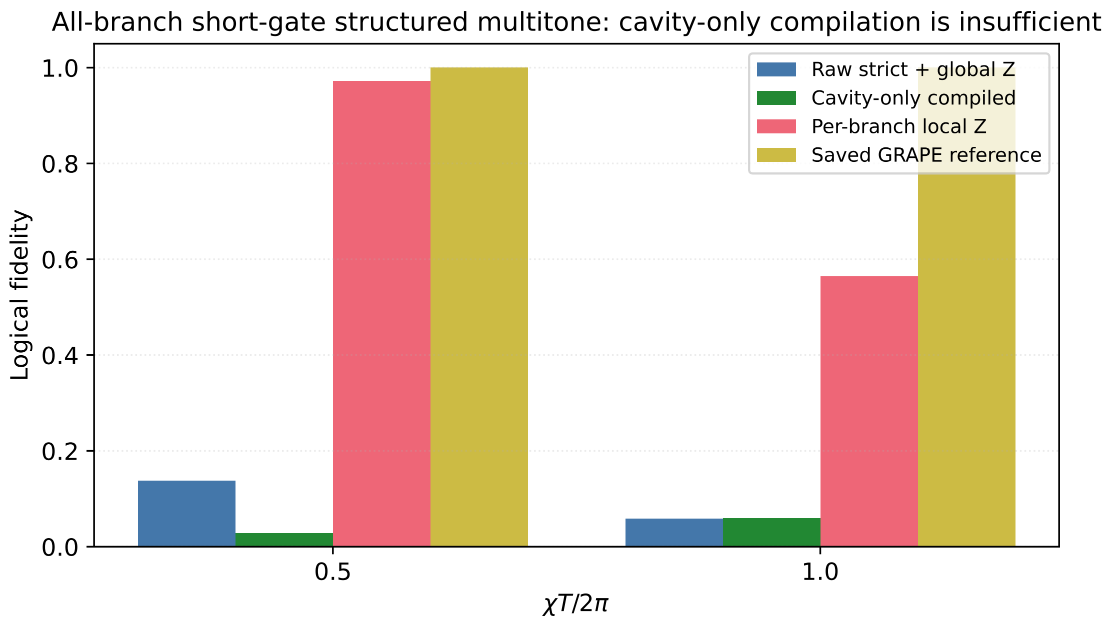

Figure 6: Follow-up all-branch short-gate comparison. Cavity-only bosonic phase compilation does not rescue the strict logical action, while branch-local-Z freedom and GRAPE reveal a much larger reachable control set.

Primitive	Nominal fidelity	Strict logical fidelity	Robustness	Hardware realism	Calibration complexity	Recommendation
Single-tone cosine-squared SQR	≈ 1.000 conditional-phase / branch-local-Z relaxed near $\chi T/2\pi = 2$–3	High on the original $N = 4$ window; 0.931 on enlarged $N = 8$ window at $\chi T/2\pi = 3$	High in the original study	High	Low	Best immediate single-target primitive
Cosine-squared SQR + linear cavity compilation	Same pulse plus fitted $\phi(n) \approx -0.068n$ on enlarged window	0.954 on enlarged $N = 8$ window at $\chi T/2\pi = 3$	High	High	Medium	Best immediate compiled upgrade when inter-Fock coherence matters
Optimized single-branch multitone	Helpful mainly at very short gate times; marginal gain for $\chi T/2\pi \geq 3$	Similar to smooth single-tone in the practical regime	Medium	Medium	Medium	Secondary option; not a practical-regime winner
Structured all-branch multitone + cavity compilation	Can look good under branch-local/block gauges	Fails badly in representative short-gate cases (0.028 at $\chi T/2\pi = 0.5$)	Unknown	Medium	Medium-high	Do not deploy as a bosonic-compiled primitive
GRAPE pulse	≈ 1.0	≈ 1.0	Not yet fully mapped here	Medium	High	Long-term benchmark / control target

Table 1: Unified recommendation table for the reproduced original study and the new phase-compilation follow-up.

3. Do not interpret poor short-gate structured all-branch performance as evidence that "only a SNAP layer is missing." In that regime the control limitation is deeper.

6 Open Problems

The merged report sharpens the next-step agenda:

- Can the residual spectator branch-local qubit-Z structure be reduced by echoed or two-layer selective pulses without sacrificing the favorable cavity-phase behavior identified here?
- Which structured waveform family closes the large gap between the current all-branch ansatz and GRAPE while remaining hardware-reasonable and calibratable?
- How do the enlarged-window compiled results change under the more realistic hardware-transfer and expanded noise models proposed in the follow-up plan?

7 Conclusion

The original study and the new follow-up now fit together cleanly. The baseline report established that smooth conditional-phase SQR is already strong in the practical regime and that GRAPE provides a much higher ceiling than the current structured ansätze. The follow-up showed that a low-order cavity-only phase layer is genuinely useful for single-target SQR on larger logical windows, but that the hardest structured short-gate failures are not bosonic-phase problems.

The best immediate experiment-facing primitive is therefore not "full GRAPE everywhere," but *cosine-squared single-target SQR, optionally compiled with a low-order cavity-only phase layer when coherence across a larger logical window matters.* The most important unresolved bottleneck is the residual branch-local qubit-phase structure that remains after that bosonic compilation step.